# ASSIGNMENT

### Introduction to Symmetric Cryptography

Cryptography and Network Security

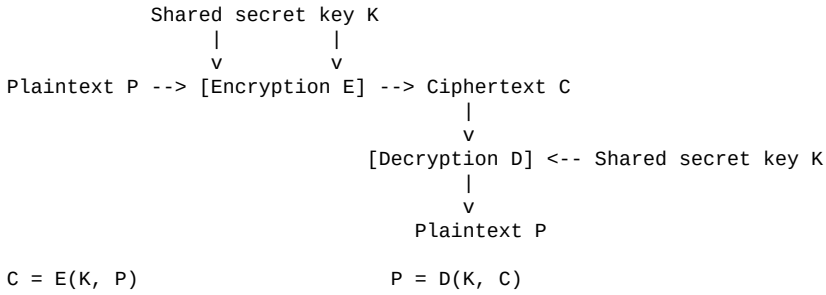
## Q1. Principles of symmetric cryptography; components; Kerckhoffs's principle

Symmetric cryptography uses one shared secret key for both encryption and decryption. The sender applies an encryption algorithm to plaintext with the secret key to produce ciphertext. The receiver applies the corresponding decryption algorithm with the same key to recover the plaintext.

• Plaintext: the original readable message or data.
• Ciphertext: the scrambled output produced by encryption.
• Secret key: the confidential value controlling encryption and decryption.
• Encryption algorithm: transforms plaintext into ciphertext using the key.
• Decryption algorithm: transforms ciphertext back into plaintext using the key.

Kerckhoffs's principle states that a cryptosystem should remain secure even if everything about the system is public except the secret key. Security should depend on protecting the key, not hiding the algorithm.

Labelled block diagram:




## Q2. Stream ciphers, block ciphers, confusion, diffusion, and modes

A stream cipher encrypts data one bit or byte at a time using a generated keystream. A block cipher transforms fixed-size blocks; AES, for example, uses 128-bit blocks.

Confusion makes the relationship between key and ciphertext complex, commonly through nonlinear substitutions. Diffusion spreads the influence of each plaintext bit across many ciphertext bits, often through permutation and mixing.

• ECB: encrypts each block independently. Simple and parallelizable, but identical plaintext blocks produce identical ciphertext blocks and reveal patterns. Usually unsuitable for multi-block data.
• CBC: XORs each plaintext block with the previous ciphertext block before encryption; the first block uses an unpredictable IV. Hides repeated patterns better than ECB, but encryption is sequential, padding is usually required, and CBC alone does not provide integrity.
• CFB: uses block-cipher output to create a stream-like mode. Can handle smaller units without padding, but has feedback/error-propagation considerations and provides no independent authentication.
• OFB: generates a keystream from repeated encryption of an internal state. Transmission errors do not propagate through later plaintext, but an IV must never be reused with the same key.
• CTR: encrypts successive counter values to create a keystream. Efficient and parallelizable, but a nonce/counter must never repeat with the same key.

These basic modes do not by themselves authenticate data. Modern applications commonly prefer authenticated encryption, such as AES-GCM, for confidentiality and integrity together.

## Q3. Historical development of symmetric encryption

Early symmetric encryption included manual substitution and transposition methods. The Caesar cipher shifted letters by a fixed amount and was easy to use but easy to break. The Enigma machine, used by Germany during World War

II, used electromechanical rotors to create changing substitutions; its cryptanalysis demonstrated both the potential and limitations of mechanical cryptography.

Claude Shannon's work on information theory, particularly his 1949 work on secrecy systems, gave cryptography a mathematical foundation. His concepts of confusion and diffusion influenced modern cipher design.

The Feistel structure divides a block into halves and applies repeated rounds using round keys. It influenced the design of Lucifer and DES. In 1977, the U.S. National Bureau of Standards adopted DES as a data encryption standard. DES became influential, but its effective 56-bit key eventually became too short against increasing computing power.

Triple DES (3DES) applied DES multiple times to improve security while retaining compatibility, but it was slower than newer designs. Other algorithms included IDEA and Blowfish (block ciphers) and RC4 (a stream cipher). RC4 later became unsuitable for secure use because of weaknesses and should not be used in modern protocols.

As computing power and cryptanalysis advanced, NIST organized an open competition to replace DES. Rijndael was selected and became the Advanced Encryption Standard (AES), published in 2001. AES supports 128-, 192-, and 256-bit keys and is widely used today.

## Q4. Classical substitution ciphers and frequency analysis

(a) Caesar cipher: each letter is shifted by a fixed number of alphabet positions. With a shift of 3, HELLO becomes KHOOR. It is simple, but has only 25 nontrivial shifts in English and is easy to brute-force.

(b) Monoalphabetic substitution: each plaintext letter is consistently replaced by another letter. Its key space is larger than Caesar's, but each letter's frequency pattern is preserved, making frequency analysis effective.

(c) Playfair cipher: encrypts pairs of letters using a 5×5 key square, usually combining I and J. Letters in the same row are replaced by letters to their right; letters in the same column by letters below; otherwise, rectangle-corner rules are used. It hides single-letter frequencies better than simple substitution, but digraph patterns and other statistical clues can still be exploited.

(d) Vigenère cipher: a repeating keyword selects different Caesar shifts for successive letters. It reduces the effectiveness of simple single-letter frequency analysis, but repeating-key patterns can reveal key length; then frequency analysis can be applied to each position group.

Frequency analysis is effective when a cipher preserves stable letter frequencies, as Caesar and monoalphabetic substitution do. It is less directly effective against digraph and polyalphabetic ciphers, but classical versions are not secure against modern cryptanalysis.

## Q5. Transposition, product ciphers, and one-time pad

A transposition cipher rearranges character positions without changing the characters themselves.

Rail Fence: write plaintext diagonally across a chosen number of rails and read row by row. It is simple but preserves letter frequencies and can be attacked by trying likely rail counts.

Columnar transposition: write plaintext in rows under a keyword, then read columns in the order determined by the keyword's alphabetical order. It creates more complex rearrangements, but preserves letter frequencies and can be attacked using structural clues and trial.

A product cipher combines substitution and transposition in multiple stages. Substitution changes symbols, while transposition redistributes their positions. Repeated rounds create confusion and diffusion; related design ideas form the basis of modern block ciphers, implemented with carefully analyzed transformations and key schedules.

A one-time pad combines plaintext with a truly random key as long as the message, used only once and kept secret. Under these conditions it offers perfect secrecy in theory. It is impractical for many uses because generating, distributing, storing and managing a random key as long as every message is difficult. Reusing a pad or using a predictable key destroys this property.

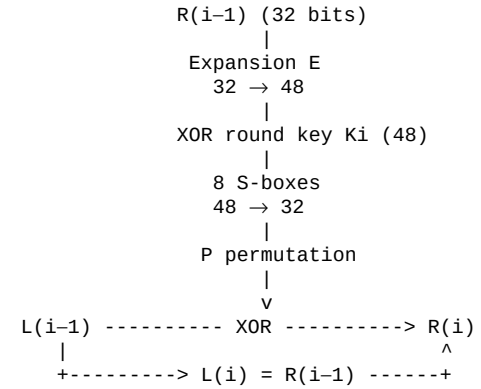
## Q6. DES structure and one round

DES is a symmetric block cipher processing 64-bit blocks. Its key is commonly described as 64 bits, but 8 bits are parity bits, leaving an effective 56-bit key.

1. Initial permutation (IP): rearranges the 64 input bits according to a fixed table.
2. Split: divides the result into 32-bit halves, L0 and R0.
3. Sixteen Feistel rounds: each new left half is the previous right half; each new right half is the previous left half XOR the round function applied to the previous right half and round key.
4. Final arrangement and inverse initial permutation (IP–1) produce the ciphertext.

The round function expands the 32-bit right half to 48 bits, XORs it with a 48-bit round key, passes it through eight S-boxes (48 bits to 32 bits), then applies a fixed P-box permutation.

One DES round:




## Q7. DES strengths, weaknesses, Double DES and Triple DES

DES has a Feistel structure and an avalanche effect: changing one input or key bit should change many ciphertext bits, on average about half. Its S-boxes and round structure were designed to resist known cryptanalytic methods, but its main practical weakness is its 56-bit effective key.

Brute-force attacks try candidate keys; 2^56 possibilities no longer provide an adequate modern security margin. Differential cryptanalysis studies how input differences affect output differences, while linear cryptanalysis uses statistical relations between plaintext, ciphertext and key bits.

Double DES encrypts twice: C = E(K2, E(K1, P)). Although it appears to provide a 112-bit key space, a meet-in-the-middle attack computes E(K1, P) for possible K1 values and D(K2, C) for possible K2 values, then searches for matching intermediate values. This reduces the work compared with trying all key pairs.

Triple DES applies DES three times, commonly using encrypt-decrypt-encrypt (EDE). Two-key 3DES uses K1, K2, K1; three-key 3DES uses independent K1, K2 and K3. It improved security over single DES and preserved compatibility, but is slow and retains a 64-bit block size. Triple DES is deprecated or disallowed for many new applications; AES is preferred today.

## Q8. AES transformations, key expansion, decryption and comparison with DES

AES is a symmetric block cipher standardized by NIST after an open competition. It uses a fixed 128-bit block size and supports key lengths of 128, 192 and 256 bits, requiring 10, 12 and 14 rounds respectively.

The 128-bit block is represented as a 4×4 byte state. Its transformations are:
• SubBytes: replaces each byte using a nonlinear S-box.
• ShiftRows: cyclically shifts the rows by different offsets.
• MixColumns: mathematically mixes bytes within each column.
• AddRoundKey: XORs the state with a round key.

Encryption begins with an initial AddRoundKey. Main rounds apply SubBytes, ShiftRows, MixColumns and AddRoundKey; the final round omits MixColumns. Key expansion derives round keys from the original key using operations including byte substitution, word rotation and round constants.

Decryption reverses encryption with inverse transformations: InvShiftRows, InvSubBytes, AddRoundKey and InvMixColumns in the appropriate order, using expanded round keys in reverse order.

Comparison:
• Structure: DES uses a 16-round Feistel network and a 64-bit block; AES uses a substitution-permutation network and a 128-bit block.
• Security: DES has 56 effective key bits; AES supports 128-, 192- and 256-bit keys and provides a much larger security margin.
• Performance: AES is generally faster in modern software and hardware; 3DES is comparatively slow.
• Current status: DES is obsolete for secure use; AES is widely adopted. Security also depends on a suitable mode, correct nonce/key management, and authentication where required.